# Tutorial 3

## Task 1

### Dataset & Preprocessing

- Dataset: MNIST, 28x28 grayscale images of digits 0–9.
- Train/Val/Test split: 48,000 / 12,000 / 10,000 images.
- Images converted to tensors (pixel values scaled to [0, 1]) using transforms.ToTensor().
- Batch size: 32.

### Model Architecture

A fully connected network (MNISTNet) with two hidden layers:

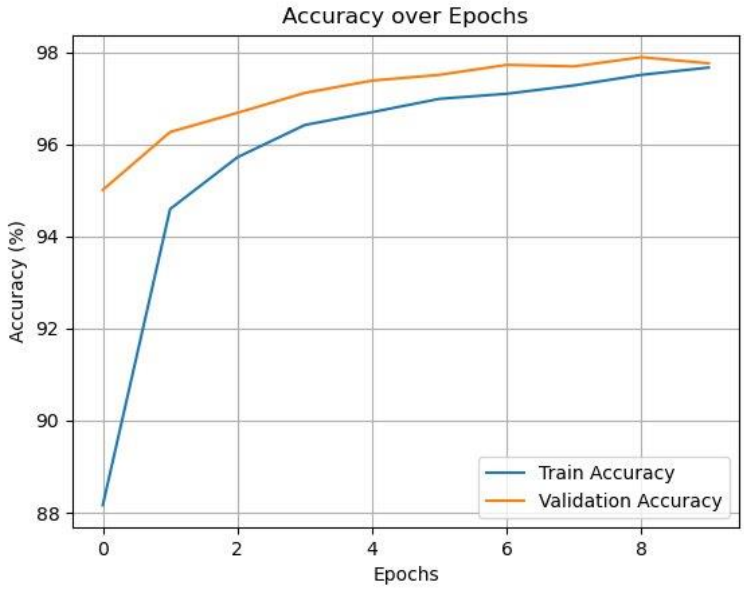
- Input: Flatten 28x28 → 784 features
- FC1: Linear(784, 128) → ReLU → Dropout(0.2)
- FC2: Linear(128, 64) → ReLU → Dropout(0.2)
- Output: Linear(64, 10) — raw logits for 10 classes

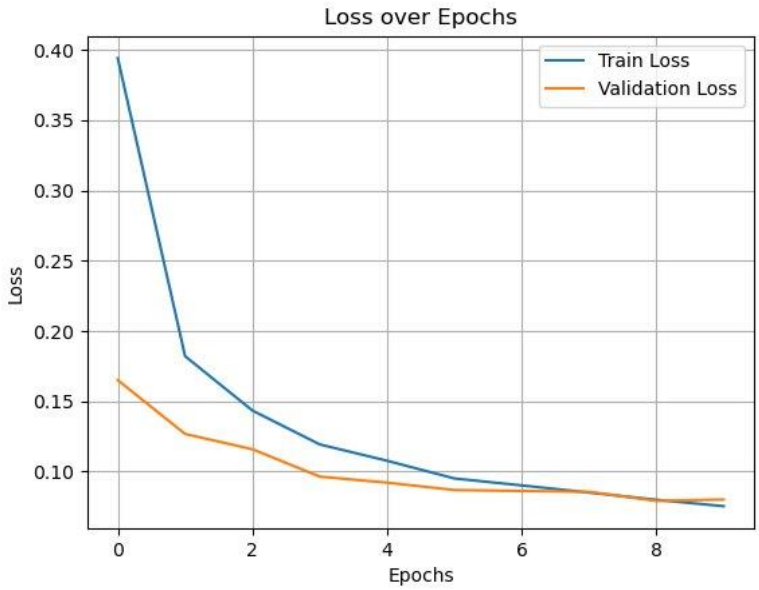
### Training Setup

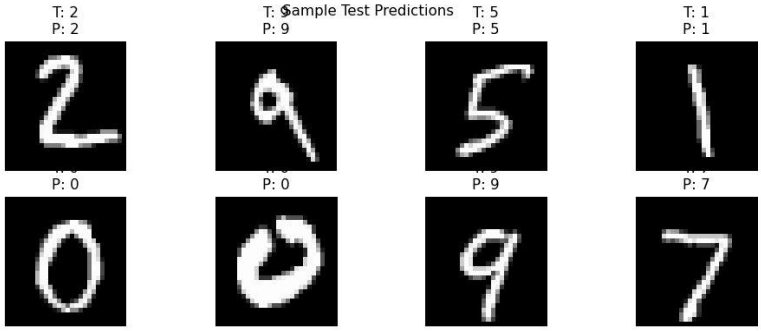
- Loss: CrossEntropyLoss
- Optimizer: Adam (lr=0.001, weight_decay=1e-4)
- Epochs: 10, Batch size: 32

### Results

- Train Accuracy: ~97.7% , Validation Accuracy: ~97.8% ,Test Accuracy: ~97–98%
- Training and validation loss both decreased smoothly with no divergence.
- Model is well-fitted , no overfitting, no underfitting.
- Dropout + Weight Decay kept train and validation curves tightly aligned.


Accuracy over Epochs
Accuracy (%)
Epochs
Train Accuracy
Validation Accuracy


Loss over Epochs
Loss
Epochs
Train Loss
Validation Loss


Sample Test Predictions
T: 2
P: 2
T: 9
P: 9
T: 5
P: 5
T: 1
P: 1
P: 0
P: 0
P: 9
P: 7

## Task 2

### Modifications Tested

- Extra hidden layer (784→128→64→32→10): Marginal gain; MNIST is simple enough that 2 layers suffice.
- Wider layers (784→256→128→10): Slightly faster convergence, similar final accuracy (~98%).
- tanh activation: Stable training, ~96–97% accuracy, slightly below ReLU.
- sigmoid activation: Slower convergence due to vanishing gradients, ~95–96% accuracy.

### Conclusion

ReLU outperforms tanh and sigmoid on MNIST. The original two-layer architecture is already well-optimised. Adding depth or width gives diminishing returns for this dataset.

## Task 3

### Results Summary

- SGD (lr=0.01, momentum=0.9): Slowest convergence (~83% at Epoch 1), reaches ~96–97% by Epoch 10.
- RMSprop (lr=0.001): Faster than SGD, reaches ~95% by Epoch 3, final ~97–98%.
- Adam (lr=0.001): Fastest — ~95% by Epoch 1, final ~97.7%. Best overall choice.

### Conclusion

All three optimizers reach similar final accuracy (~97%), but Adam converges the fastest and requires the least tuning. It is the recommended optimizer for this task.

## Task 4

### Model Fit Diagnosis

- Well-fitted: Train (~97.7%) and Val (~97.8%) accuracy are close and high. Loss curves converge without divergence.
- No overfitting: Validation loss never rises. Val accuracy is not falling behind train accuracy.
- No underfitting: Both accuracies are high from early on.

### Effect of Epochs

- Too few epochs this would cause underfitting (high loss, low accuracy).
- Too many epochs this would cause overfitting risk (val loss rises while train loss falls).
- 10 epochs is appropriate here given the regularisation applied.

### Early Stopping

Monitor validation loss each epoch. If it does not improve for a set number of epochs (patience), stop training and restore the best weights. For this model, early stopping would trigger around Epoch 8–9 as val loss plateaus.

## Regularisation Techniques

- Dropout (0.2): Randomly disables 20% of neurons per pass, forces the model to learn distributed representations, reducing overfitting.
- Weight Decay (1e-4): L2 penalty on large weights, smooths the learned function and improves generalisation.
- Combined effect: Training and validation curves stay tightly aligned throughout all 10 epochs.

## Conclusion

The MNISTNet model achieves **~97–98%** test accuracy using a two-layer fully connected network with ReLU activations, Dropout, and Adam optimisation. Architecture experiments confirmed the original design is efficient for this task. Adam was the best optimizer tested. The learning curves confirm a well-fitted model with effective regularisation throughout training.